

Heterogeneous urban decarbonization pathways revealed by typological and machine learning frameworks: A global analysis across Annex I cities

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ABSTRACT

Cities are central to global climate mitigation, yet existing projections often rely on coarse-grained assessment models that overlook the interplay between urban typology and socioeconomic dynamics. Here, we harness a context-aware deep learning framework combined with typological clustering to project CO₂ emissions for 46,833 cities in Annex I countries through 2050 under Shared Socioeconomic Pathways. Specifically, we classify cities by multidimensional urban morphology and couple the resulting typologies with FiLMSeq2Seq prediction model we proposed to generate city-level emission projections. We identify five distinct urban typologies characterized by differing patterns of horizontal expansion, economic intensification, and vertical structure. We find that emission trajectories diverge significantly across these typologies: Under the Business-As-Usual scenario, 'Sprawling' cities have the highest proportion of emission increases, with 31.5% of the 1335 cities showing growth. Conversely, 'Moderate vertical' cities will dominate future mitigation with a total emission reduction of 798.2 Mt. Our results suggest that uniform national policies are unsuitable for diverse urban typologies, underscoring the need for type-specific strategies to realize the full decarbonization potential of cities.

1. Introduction

Cities, currently home to over half of the global population, serve as the primary engines of economic growth and the dominant sources of anthropogenic carbon dioxide (CO₂) emissions (Nations, 2025; UN-HABITAT, 2025). As nations strive to limit global warming to within 1.5 °C or 2 °C, cities have emerged as the frontlines of climate mitigation efforts (Gurney et al., 2022; Sharifi et al., 2026). Consequently, accurate projection of urban CO₂ emissions is critical for assessing the feasibility of climate targets and formulating effective local decarbonization roadmaps (Hsu et al., 2022; Sharifi et al., 2025). Over the past two decades, while many Annex I countries have initiated emission reduction efforts (Grassi et al., 2022), the efficacy of these efforts is increasingly complicated by significant transformations in urbanization patterns, characterized by simultaneous outward expansion and upward vertical growth (Lwasa et al., 2022; Mahtta et al., 2019).

Mounting evidence indicates that urban emissions are driven by the complex interplay of socioeconomic factors, with key determinants including population size, GDP, and land use (Crippa et al., 2021; Franco et al., 2023; Jin and Sharifi, 2025; Luqman et al.,

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2023). Existing research typically relies on Integrated Assessment Models (IAMs) and Shared Socioeconomic Pathways (SSPs) to project future emission trajectories, operating under the assumption that shifts in these macro-variables will yield predictable changes in emissions (Grassi et al., 2021; Gurney et al., 2022; Ou et al., 2021; van Vuuren et al., 2017). These projections generally serve as benchmarks for estimating the social cost of carbon and for guiding national climate policies (Roelfsema et al., 2020).

In the domain of urban CO₂ emission projection, most studies focus on individual cities or cities within a single nation (Jin and Sharifi, 2025). While some studies have addressed urban emissions across multiple countries, they are generally limited to assessing current emissions rather than projecting future trends (Franco et al., 2022; Park et al., 2025). Furthermore, numerous studies highlight that statistical projection methods rely heavily on assumptions, whereas machine learning approaches often lack interpretability (Eyring et al., 2024; Imrie et al., 2023; Jin et al., 2024). These issues underscore the substantial uncertainty remaining in urban emission projection. Specifically, this uncertainty stems from three critical factors. First, there is a lack of officially released, consistent city-level source data (Brown and Macaskill, 2025; Hsu et al., 2020). Second, there is an absence of comparable future emission scenarios at the city level. Most prior studies (Wu et al., 2023; X. Y. Zhou et al., 2022a) established unique scenarios based on specific national and urban policies rather than a unified framework, hindering cross-comparison (Zhang and Hanaoka, 2022). Third, there is a shortage of objective and interpretable projection models capable of explaining projection results, whether through city typology or cross-national differences (Frolking et al., 2024; Mahtta et al., 2019).

Due to these challenges, few studies to date have conducted multi-level analyses of urban-national-continental aggregate CO₂ emissions (Hsu et al., 2022; Jin and Sharifi, 2025). To our knowledge, no study has projected future trends and spatial distributions for numerous cities under multiple climate scenarios. Without such granular quantification, current projections of urban decarbonization potential may be subject to significant bias (Gurney et al., 2021; Isik et al., 2021; Salvia et al., 2021).

To address these gaps, we project CO₂ emissions for 46,833 cities across 42 Annex I countries through 2050 while accounting for urban morphological heterogeneity (Fig. 1). We ask how cities can be classified by historical development patterns, how their emissions diverge under SSP1, SSP5, and BAU, and which socioeconomic drivers explain these differences. We hypothesize that emission trajectories diverge systematically by urban morphological typology, independent of national-level trends, such that cities within the same country but of different types may follow divergent decarbonization pathways. If confirmed, this would imply that uniform, nationally uniform mitigation targets may misrepresent the mitigation burden or potential of individual cities, underscoring the need for typology-informed, city-level strategies rather than top-down national benchmarks. Finally, we use FiLMSeq2Seq prediction model to generate city-level future predictions.

2. Materials and methods

Our framework comprises three components (Fig. 1): clustering cities by observed changes in urban form, projecting city-level

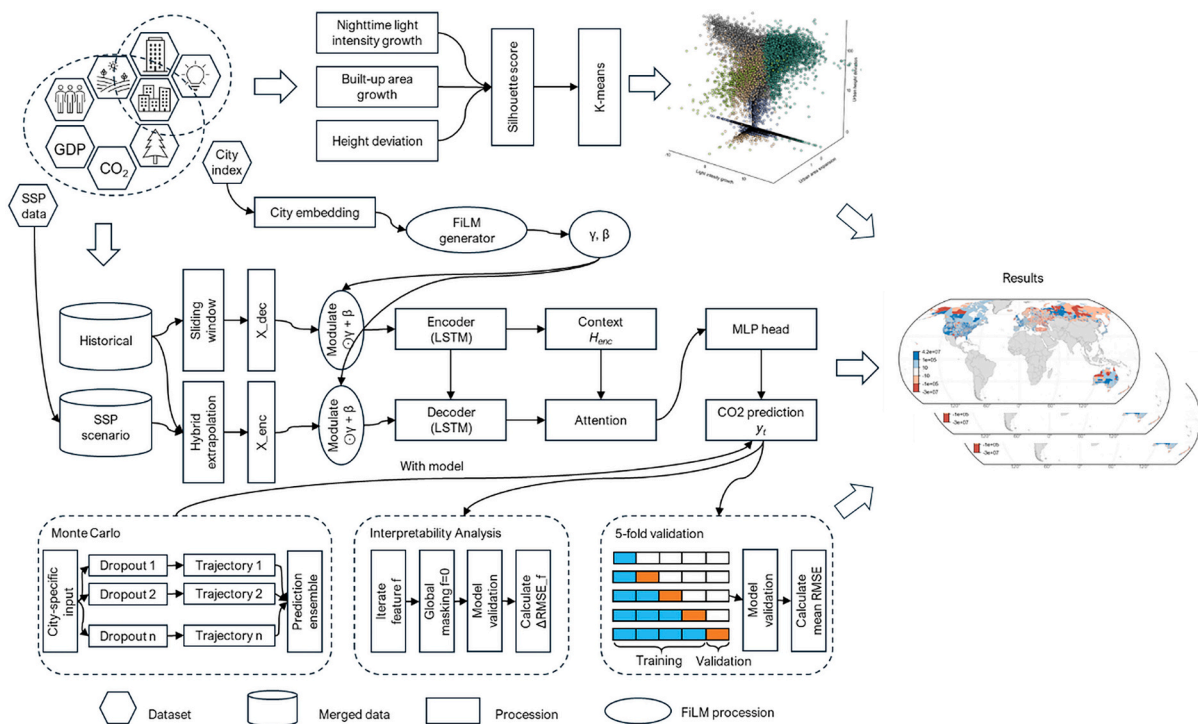


Fig. 1. Research methodology.

Overview of K-means classification and FiLMSeq2Seq projection methods.

emissions with the context-aware FiLMSeq2Seq model, and linking the projections to SSP1, SSP5, and BAU. Monte Carlo Dropout and ablation analyses are used to quantify uncertainty and assess feature sensitivity.

2.1. Dataset preparation

2.1.1. Boundary data

Boundary data were derived from the Database of Global Administrative Areas (GADM) (GADM, 2018). To determine the spatial boundaries of each city, consistent with previous studies (Hsu et al., 2022), we selected lower-level administrative divisions below the provincial, regional, or state level. For countries with multiple tiers of lower-level administrative divisions, we selected the largest tier that met the definition and merged smaller areas using Geopandas.

2.1.2. Emission data

To ensure the interpretability of our deep learning framework and the reliability of morphological clustering, we restricted our training domain to Annex I nations where consistent and continuous historical data (2001–2020) is available. Urban CO₂ emissions were calculated based on the Annex I national-level CO₂ (without LULUCF) time-series data published by the UNFCCC (Kuyper et al., 2018), combined with calibrated and simulated 500-m nighttime light grid data (Chen et al., 2024). Specifically, we first read the pixels within each city's boundaries using rasterio and calculated the sum using rasterstats to obtain the total urban Nighttime Light (NTL) intensity. National emissions were then downscaled based on the ratio of the city's total NTL intensity to the national total. This spatial allocation methodology follows the established assumption that the intensity of nighttime lights serves as an effective proxy for CO₂ emissions, a technique widely utilized in global emission inventories (Oda and Maksyutov, 2011).

2.1.3. Clustering features

The selection of clustering features was informed by the IPCC AR6 and prior research (Lwasa et al., 2022; Mahtta et al., 2019). To assess the multidimensional characteristics of cities, we augmented traditional outward-expansion and vertical-structure metrics with average urban light intensity and changes in luminosity. The source data included MODIS 500 m Type 13 built-up area, 2015 building height data (Y. Y. Zhou et al., 2022b), and mean of urban NTL. To account for dynamic urban development, we calculated the difference in built-up area between 2020 and 2001 to represent built-up area growth. For light intensity, we first computed the mean value to eliminate the influence of city size, and then calculated the difference between 2020 and 2001. The standard deviation of building heights in 2015 was calculated to represent urban height deviation.

2.1.4. Projection features

To construct a city-level time-series dataset for training and projection, we integrated historical observational data from 2001 to 2020. Based on SSP scenarios and previous studies on urban emission projection, we identified five projection features: GDP, population, and three types of land use. For GDP, we utilized the global annual gridded GDP data provided by Kummu et al. (2025) at a resolution of 30 arc-seconds. For population, we employed the Gridded Population of the World (GPW) dataset at a resolution of 30 arc-seconds (Doxsey-Whitfield et al., 2015). As this dataset provides population estimates in five-year increments from 2000 to 2020, we derived annual population estimates by linearly interpolating between these five-year intervals. Land use data were sourced from the MODIS 500-m resolution dataset (Friedl and Sulla-Menashe, 2022), covering the period from 2001 to 2020. These five variables constitute the only inputs to the FiLMSeq2Seq prediction model.

It is important to note that the projection features used in the training phase must align with those available for future scenario projections. The SSP models include four land use categories: built-up area, cropland, forest, and pasture. Since the MODIS data do not include a pasture category, we only considered the first three types. Cropland corresponds to MODIS Type 12, and forest corresponds to the sum of Types 1 through 5.

2.2. Clustering

2.2.1. Data preprocessing

Given the presence of heavy-tailed distributions and potential negative values in the raw metrics (e.g., population shrinkage or dimming lights in some cities), we applied a Signed Log Transformation (slog) to all feature vectors $x \in \mathbb{R}^N$ prior to clustering. Subsequently, we performed Z-score normalization on the transformed features to prevent variables with larger numerical magnitudes from dominating the distance metric calculations.

2.2.2. K-means clustering and validation

We employed the K-Means algorithm to classify cities into distinct clusters based on the standardized feature profiles (McQueen, 1967). This algorithm partitions data by minimizing the Within-Cluster Sum of Squares (WCSS) in the multi-dimensional feature space. To determine the optimal number of clusters (k), we conducted a Silhouette Analysis (Rousseeuw, 1987) (Supplementary Data Fig. 5), testing k values ranging from 2 to 10. We calculated the average silhouette coefficient for each k under the Euclidean distance metric, which measures the cohesion within clusters and the separation between them. Based on the peak silhouette coefficient and the physical interpretability of the resulting classifications, we selected $k = 5$ as the final classification scheme. All clustering procedures were implemented using Python's Scikit-learn library, with 10 random initializations to ensure the algorithm converged to a stable global optimum.

2.3. Projection model

To capture the complex spatiotemporal heterogeneity across different cities, we developed a Sequence-to-Sequence (Seq2Seq) deep learning model integrated with FiLM (Perez et al., 2018; Sutskever et al., 2014). The novelty of the model lies in combining city-specific FiLM conditioning, attention, and Monte Carlo Dropout for large-scale urban emission projection.

2.3.1. City-specific conditioning via FiLM

The core of the model lies in the FiLM layer, which dynamically modulates the neural network's computational flow by learning city-specific embeddings. For each city c , the model first maps its unique identifier to an embedding vector $e_c \in \mathbb{R}^{d_{emb}}$. A linear generating network (FiLM Generator) then produces two sets of modulation parameters based on e_c : a scaling factor γ_c and a shifting factor β_c . For input features x , the FiLM operation is defined as an affine transformation:

$$\text{FiLM}(x|c) = \gamma_c \odot x + \beta_c \quad (1)$$

where $\odot$ denotes element-wise multiplication. This mechanism allows the model to adaptively adjust its sensitivity to input projection features based on the socioeconomic attributes of different cities (implicitly encoded in e_c) while sharing global parameters.

2.3.2. Sequence modeling with attention

The backbone of the model consists of an Encoder and a Decoder, both constructed using two-layer Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997) (Fig. 1). The Encoder receives a historical sequence X_{enc} of length $L = 12$, while the Decoder receives the covariate sequence X_{dec} for the future scenario. To enhance the capability of capturing long-term dependencies, we introduced the Luong Attention Mechanism in the decoding phase (Luong et al., 2015). At each decoding time step t , the attention module calculates the relevance scores between the decoder's hidden state h_t^{dec} and all encoder hidden states H^{enc} to generate a context vector c_t , thereby enabling the model to focus on the most projection temporal segments within the historical window.

2.3.3. Training parameters

The model was implemented using the PyTorch framework (Paszke et al., 2019). The loss function employed was Mean Squared Error (MSE), and parameters were updated using the Adam optimizer with an initial learning rate of 8×10^{-4} and weight decay of 10^{-5} to prevent overfitting. We adopted a learning rate decay strategy (ReduceLROnPlateau), halving the learning rate if the validation loss did not decrease for 5 epochs. The model was trained for 50 epochs with a batch size of 128. To ensure training stability, we applied Gradient Clipping (norm = 1.0). The final model selection was based on Early Stopping using the Root Mean Squared Error (RMSE) on the validation set.

2.4. Scenario construction

For future scenarios extending from 2021 to 2050, we designed two projection feature extrapolation strategies to capture different socioeconomic trajectories.

2.4.1. BAU scenario

For the BAU scenario, we assumed that future urban development would follow the specific momentum observed over the past two decades. For each city c and feature v , we fitted a linear trend line using Ordinary Least Squares (OLS) regression on historical projection features (2001–2020) to derive the historical growth rate, denoted as $\beta_{c,v}$. Future projection features (2021–2050) were then projected from the base year ($t_{base}=2020$):

$$\hat{x}_{c,t}^{lin} = \max(0, x_{c,t_{base}} + \beta_{c,v} \cdot (t - t_{base})) \quad (2)$$

where the $\max(0, \cdot)$ function enforces non-negativity constraints to prevent physically unrealistic negative values (e.g., for population or built-up areas) in shrinking cities.

2.4.2. SSP scenarios

For the SSP1 and SSP5 scenarios, we implemented a Hybrid Extrapolation Mechanism designed to reconcile local historical inertia with macro-regional developmental pathways. First, we collected the SSP projection values for five projection features from 2020 to 2050 via the IIASA SSP Database (Riahi et al., 2017). We derived the annual estimates for 2021–2050 by performing linear interpolation on these scenario data. Next, the city-level SSP future scenarios were generated by calculating the weighted average of the local linear extrapolations and the SSP trajectories. Finally, the 2021–2050 future scenario features for BAU and SSPs were input into the trained projection model to project future urban CO₂ emissions:

$$\hat{x}_{c,t}^{SSP} = w \cdot \hat{x}_{c,t}^{lin} + (1 - w) \cdot \left(x_{c,t_{base}} \times \frac{S_t}{S_{t_{base}}} \right) \quad (3)$$

where $\hat{x}_{c,t}^{\text{lin}}$ is the value derived from the city-specific linear extrapolation described above. S_t represents the corresponding variable from the SSP database at time t . w is the weighting factor, set to 0.5 in this study to balance urban historical continuity with macro SSP scenarios. All input features underwent city-level standardization (Z-score normalization) before being input to the model. SSP1 and SSP5 represent contrasting low- and high-emission development pathways, while BAU extrapolates historical trends without additional scenario assumptions. Together, they span a plausible range of future urban development and provide a reference for evaluating continuation of current trends.

2.5. Uncertainty analysis

Given the inherent uncertainty in long-term climate projections, we employed Monte Carlo Dropout as a Bayesian approximation inference method (Gal and Ghahramani, 2016). During the training phase, we applied Dropout with a probability of $p = 0.1$ after the LSTM and fully connected layers. Unlike conventional inference, we kept Dropout active during the projection phase and performed $N = 20$ stochastic forward passes for each city-scenario pair.

For any projection time t , the model's final projection distribution is described by the statistics of these N samples. We calculated the sample mean as the deterministic projection result, and computed the sample standard deviation along with the 5% and 95% quantiles to construct a 90% projection confidence interval. This enables the model to output not only point estimates but also to quantitatively assess the projection confidence under different scenario assumptions.

2.6. Interpretability analysis

To decouple the contribution of individual socioeconomic projection features to the model's projection performance and to verify that the learned feature representations align with physical laws, we conducted a systematic ablation analysis on the validation set. This post-hoc interpretability method quantifies feature importance by suppressing specific input information and measuring the resulting degradation in projection accuracy.

We trained a separate FiLMSeq2Seq model instance fitted exclusively on historical data from 2001 to 2020. Feature importance evaluation was strictly performed on the Validation Set to ensure the analysis reflected the model's generalization logic rather than rote memorization of training data. We employed Zero-masking as the perturbation strategy (Zeiler and Fergus, 2014). Since all input features were standardized (mean 0, variance 1) prior to training, masking a feature involved replacing its values with 0.0 (the global training mean). This approach removes the variance of the target feature's information while preserving the central tendency of the data distribution, thereby minimizing interference artifacts introduced by out-of-distribution inputs. Feature importance was quantified by the increase in Root Mean Squared Error (ΔRMSE) (Breiman, 2001). For each covariate f (e.g., GDP, built-up area), we simultaneously occluded the feature across all time steps in both the Encoder and Decoder. The resulting performance drop represents the total projection value of that feature to the system.

2.7. Model comparison with cross validation

To demonstrate the superiority of the proposed FiLMSeq2Seq model in capturing spatiotemporal dependencies and urban heterogeneity, we used three CO₂ emission projection models from previous studies as baselines (Hsu et al., 2022; Li et al., 2024; Pottinger et al., 2024), as shown below. These four models were applied to historical data from 2001 to 2020. They were compared using a 5-fold expanding window time-series cross-validation (Fig. 1). The results in Supplementary Data Fig. 6 verify the superiority of the proposed model.

2.7.1. LSTM

To quantify the specific contributions of the FiLM and attention mechanisms in our proposed architecture, we adopted a standard LSTM as a neural benchmark (Hochreiter and Schmidhuber, 1997). The LSTM employs a simpler architecture: historical features are concatenated with city embeddings and processed directly through a multi-layer LSTM encoder. To ensure a fair comparison, its hyperparameters were kept consistent with the proposed model: hidden layer dimension of 128, 2 layers, and a learning rate of 8×10^{-4} .

2.7.2. Extreme gradient boosting (XGBoost)

XGBoost is an efficient tree-based gradient boosting algorithm known for its scalability and high performance in regression tasks (Chen and Guestrin, 2016). It optimizes a training objective by ensembling weak projection models (typically decision trees) and employs regularization techniques to prevent overfitting. To adapt time-series data for this tree-based model, we flattened the historical time-series features (12 years) into a one-dimensional feature vector for each sample. We utilized the MultiOutputRegressor framework to simultaneously project multi-step future horizons. Based on a grid search, the optimal hyperparameters were set as follows: number of estimators ($n_{\text{estimators}}$) = 500, maximum depth (max_depth) = 6, learning rate (learning_rate) = 0.05, and subsample ratio (subsample) = 0.8.

2.7.3. Random forest (RF)

Random Forest is a well-known ensemble learning method based on bagging (Breiman, 2001). Unlike boosting methods, RF

constructs a large number of decision trees independently during training and outputs the average projection, making it highly robust to noise and less prone to overfitting in high-dimensional tabular data. Consistent with the XGBoost baseline, sequence input data were flattened into feature vectors. The model configuration included 500 estimators with a maximum depth limit of 10 to balance model complexity and generalization capability.

2.7.4. Evaluation metrics

To quantitatively assess CO₂ emission projection performance across all cities and validation folds, this study employed the following two metrics.

The Coefficient of Determination (R^2) score measures the proportion of the variance in the dependent variable that is predictable from the independent variables. An R^2 score closer to 1 indicates a higher correlation between the projected trajectory and the ground truth.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

RMSE is a standard metric for regression tasks, measuring the square root of the average squared difference between projected values ($\hat{y}_i$) and observed values (y_i). It penalizes large errors more heavily and provides a robust measure of projection accuracy in the original units of emissions.

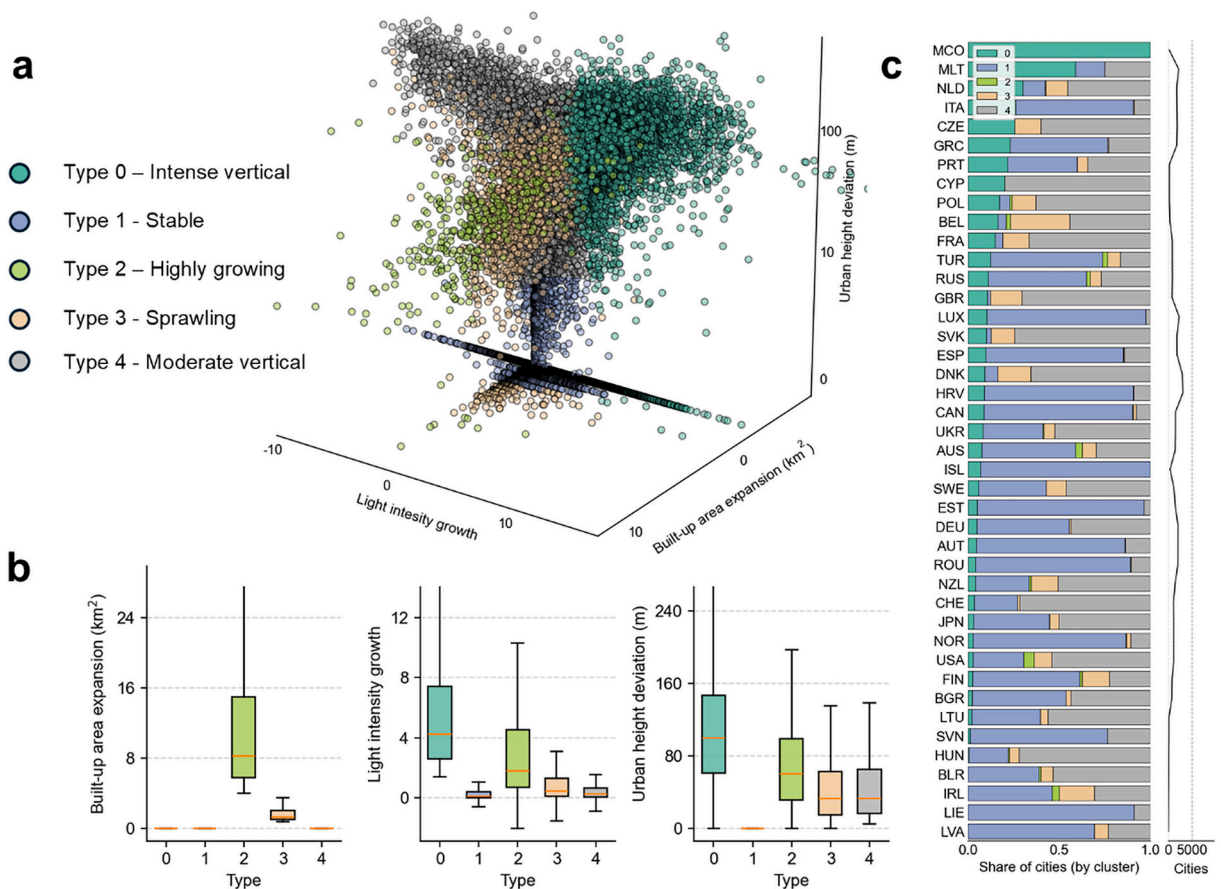


Fig. 2. Clustering distribution of Annex I cities.

a, K-means clustering of 46,833 cities based on three dimensions: built-up area expansion (2001–2020), nighttime light intensity growth (2001–2020), and built-up height deviation in 2015. **b**, Distributions of built-up area expansion, light intensity growth, and height deviation. In the box plots, the lower and upper edges of the box represent the 25th and 75th percentiles, respectively; the horizontal line inside the box indicates the median; the box height corresponds to the interquartile range (IQR); and the lower and upper whiskers represent the minimum and maximum values within 1.5 times the IQR from the 25th and 75th percentiles, respectively. **c**, The proportion of city types in 42 Annex I countries (left panel) and the total number of cities in each country (right panel).

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

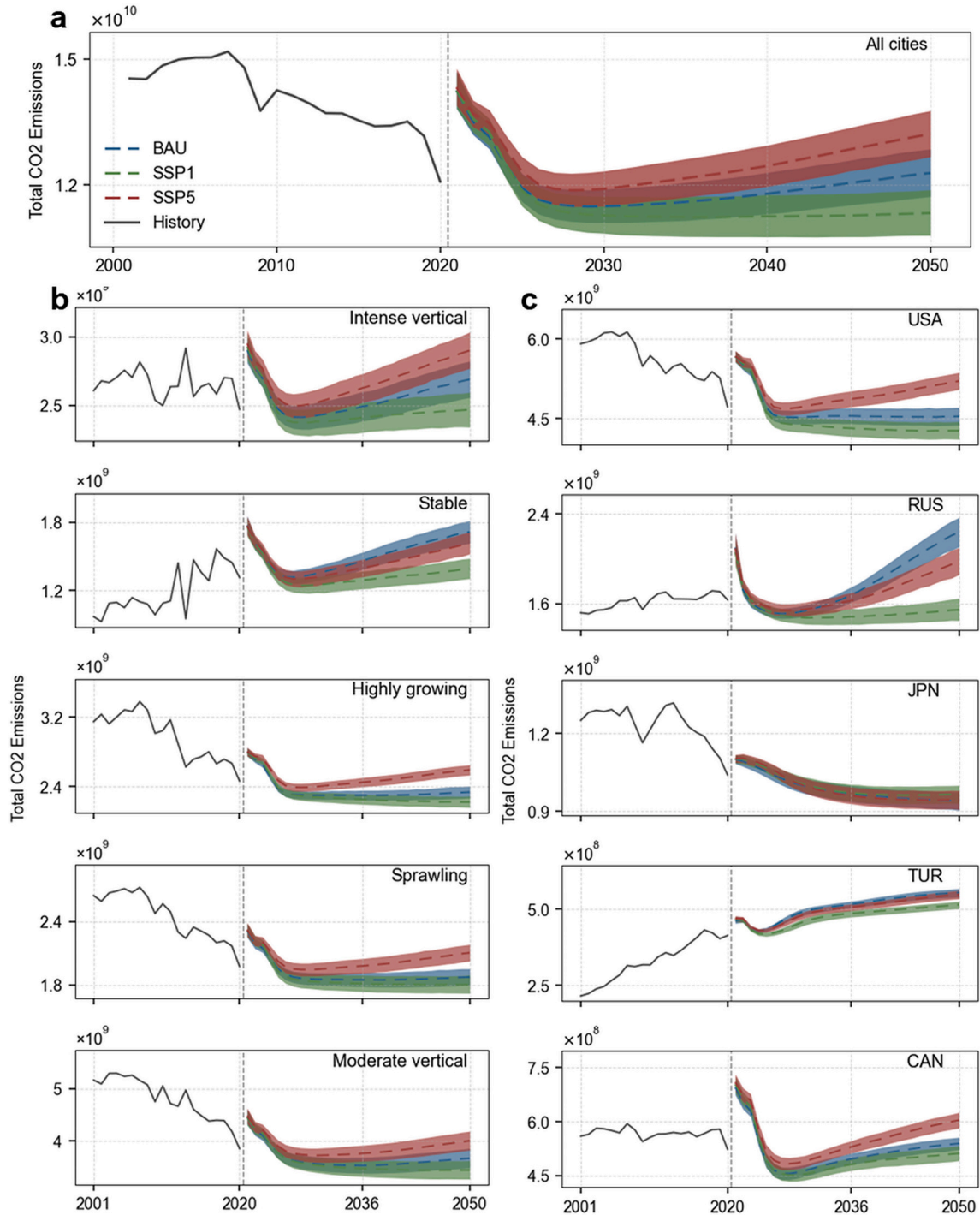


Fig. 3. CO₂ emission trajectories for all cities, city types, and major countries through 2050.

a, Solid lines represent total CO₂ emissions across all cities; historical trends (left) are shown in gray, while future projections (right) are colored by scenario, representing the mean values of Monte Carlo simulations. Translucent shaded bands denote the 5%–95% uncertainty intervals. The vertical dashed line marks the boundary between the historical and projection periods. **b,** Total CO₂ emissions disaggregated by city types. **c,** Total CO₂ emissions for the top five emitting countries under the BAU scenario in 2050.

3. Results

3.1. Classification of Annex1 cities

All Annex 1 cities were classified into five types using historical data from 2001 to 2020 (Fig. 2a). Urban height deviation reflects vertical structural heterogeneity, built-up area growth indicates horizontal expansion, and the increase in light intensity signals economic intensification. Of the 46,833 Annex I cities, over half (28,935) are categorized as ‘Stable’ cities, characterized by near-zero built-up area growth and minimal height deviation. The ‘Intense vertical’ cities (5146 cities) are characterized by pronounced vertical structural heterogeneity and economic intensification rather than outward expansion; nearly all exhibited elevated height deviation alongside nighttime light growth, while 91% showed zero horizontal growth. Similarly, the ‘Moderate vertical’ cities (11,088 cities) reflect moderate vertical structural heterogeneity, with 100% recording positive height deviation alongside modest nighttime light growth. The ‘Sprawling’ cities (1335 cities) type is defined by steady outward expansion, with all cities showing positive built-up area growth and roughly 89% also exhibiting elevated height deviation and nighttime light growth. Finally, the ‘Highly growing’ cities (329

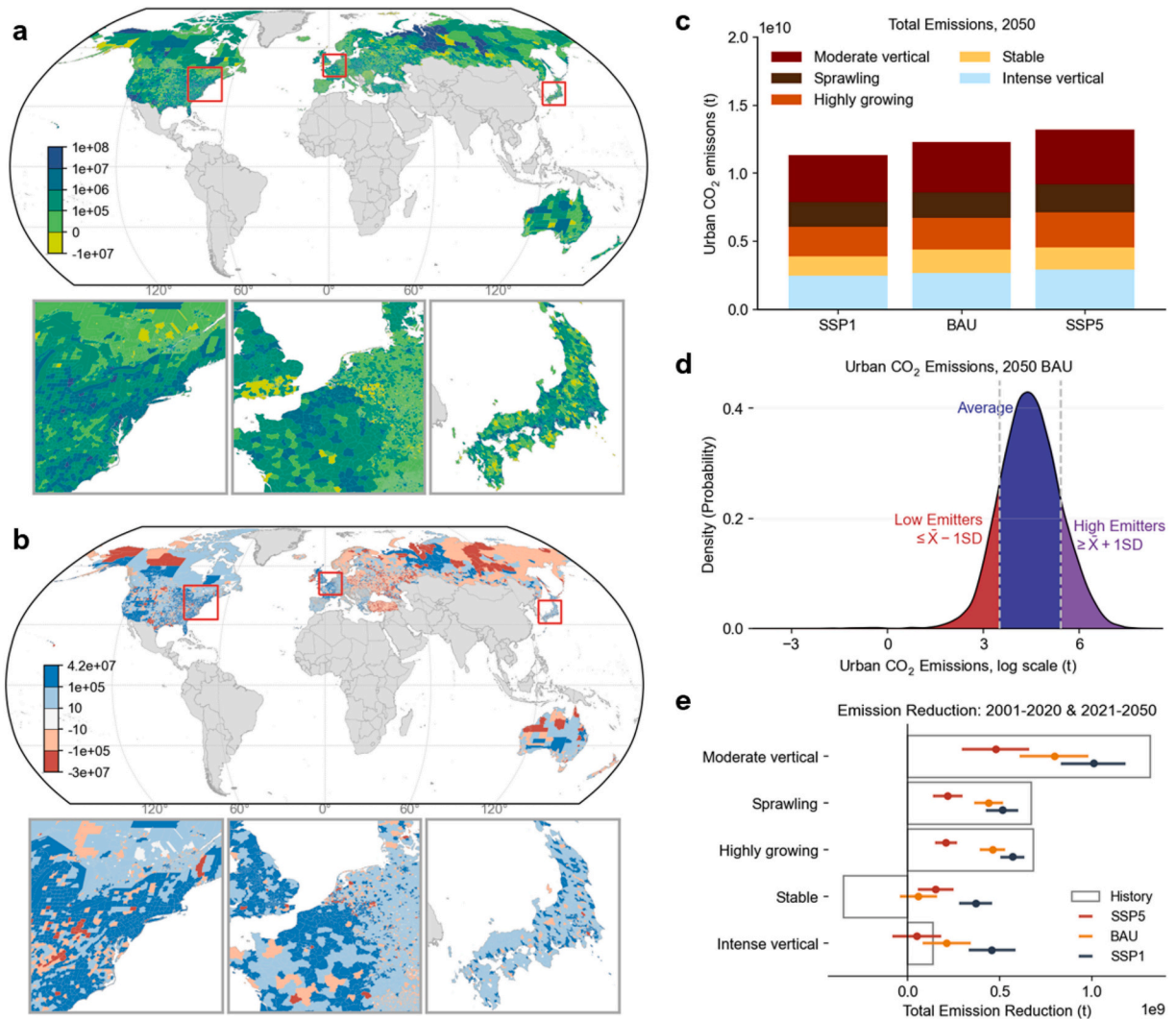


Fig. 4. Spatial distribution and scenario comparison of emissions and reductions for 46,833 Annex I cities under future scenarios. **a**, City-level CO₂ emissions in 2050 under the BAU scenario. Darker green indicates higher emissions. The three red box insets correspond to the subplots below, displaying (from left to right) parts of Eastern North America, Western Europe, and Japan. **b**, Changes in city-level CO₂ emissions from 2021 to 2050. Blue indicates a decrease in emissions, while red indicates an increase. **c**, Comparison of total emissions across three scenarios in 2050, with bars colored by city type. **d**, Density distribution of urban emissions under the 2050 BAU scenario, including only cities with emissions greater than zero. A log transformation is applied to the x-axis. High and low emitters are delineated using the mean $\pm$ standard deviation. **e**, Comparison of total historical versus future emission reductions. Historical reductions are shown as white bars, while future reductions are colored by scenario. For the error bars, dots represent the mean, and the upper and lower whiskers represent the 5%–95% uncertainty intervals.

cities) represents a small but extreme subset marked by aggressive, multi-dimensional development: 100% exhibited massive horizontal sprawl (mean growth of 15.89 km²), coupled with near-universally elevated height deviation (97.6%) and nighttime light growth (97.3%).

Highly-growing cities account for only 0.7% of the total and are primarily distributed along the coastal regions of the United States and the east coast of Australia, as well as in Türkiye (Supplementary Data Fig. 1). Meanwhile, Moderate vertical cities comprise the majority of cities in European countries such as the UK and France. Overall, Stable cities and Moderate vertical cities dominate in most countries (Fig. 2c).

From 2001 to 2020, 41,521 Annex 1 cities (88.7%) experienced an increase in light intensity, while the proportions of cities with decreasing or unchanged light intensity were 7.9% and 3.4%, respectively. Notably, all “Intense vertical” cities showed an increase in light intensity, with a mean value of 7.6, significantly higher than the overall average of 1.2 (Fig. 2b). Similarly, all Highly growing and Sprawling cities exhibited growth in built-up area, with average increases of 15.9 km² and 1.6 km², respectively. Approximately 39.4% of cities exhibited built-up height deviation, with 25% of cities having a standard deviation greater than 29.4 m.

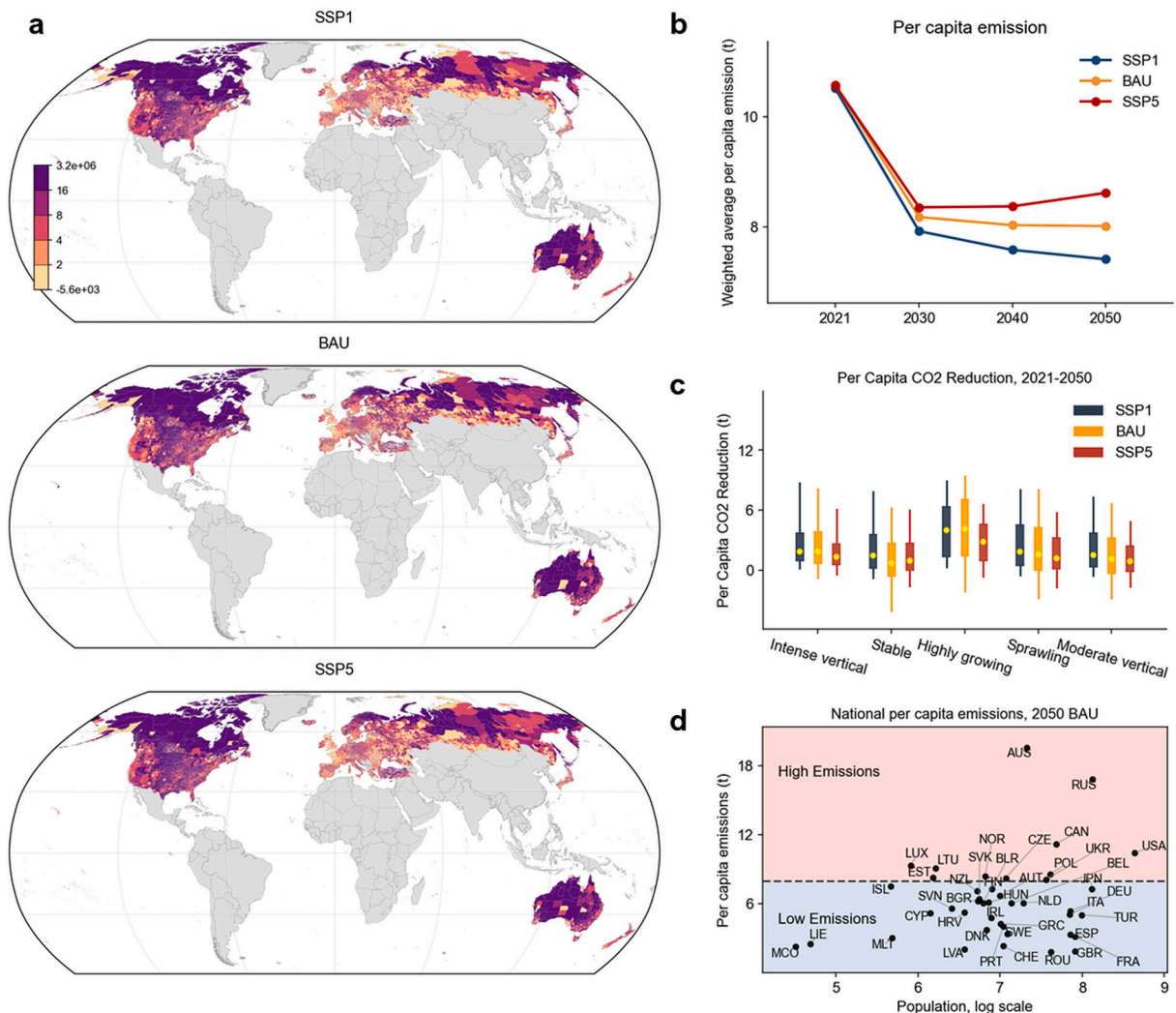


Fig. 5. Distribution of per capita CO₂ emissions for 46,833 Annex I cities under SSP scenarios.

a, Per capita CO₂ emissions in 2050. Darker red indicates higher emissions. The top, middle, and bottom panels correspond to the SSP1, BAU, and SSP5 scenarios, respectively. **b**, Trends in per capita emissions for 2020, 2030, 2040, and 2050. Data for 2020 represent historical values (shown in gray), while other years represent future projections colored by scenario. **c**, Comparison of per capita emission reductions across three scenarios in 2050, disaggregated by city type. The central dot represents the median; the lower and upper edges of the boxes represent the 25th and 75th percentiles; and the lower and upper whiskers represent the 10th and 90th percentiles. **d**, Distribution of per capita CO₂ emissions for Annex I countries under the BAU scenario in 2050. The horizontal axis represents the total population of the country on a logarithmic scale. The vertical axis shows per capita CO₂ emissions, with black whiskers indicating the population-weighted mean per capita emissions.

3.2. Future projection

Fig. 3 illustrates the temporal trends of total emissions for Annex 1 cities, city types, and countries. For all Annex 1 cities (Fig. 3a), the three scenarios show an overall declining trend from 2021 to 2050, with SSP1 exhibiting the largest decrease of 20.5%. However, varying degrees of rebound occurred between 2036 and 2050, with SSP5 showing the most significant increase of 1005.5 Mt. (8.2%). This is primarily because the emission reduction trends in most countries with declining emissions gradually slowed over time, whereas the trends in some countries with rising emissions, such as Russia, did not slow down (Supplementary Data Fig. 2).

From 2021 to 2050, Highly growing cities, Sprawling cities, and Moderate vertical cities consistently showed a declining trend under the SSP1 scenario, with decreases of 20.4%, 22.3%, and 22.7%, respectively (Fig. 3b). In contrast, Intense vertical cities and Stable cities exhibited a trend of decline followed by a rise, with increases of 2.4% and 7.7% respectively between 2036 and 2050. Fig. 3c displays the emission trends of the five countries with the highest emissions in 2050. Under the BAU scenario, Russia and Türkiye experienced the largest growth in emissions, while Japan and the United States achieved the most significant reductions (Fig. 3c).

3.3. Emission reductions

As shown in Fig. 4a and b, we analyzed 2021–2050 emission trajectories under the BAU scenario for the three cities with the largest emission increases and decreases in each country across five city types. In Intense vertical cities, such as Osaka in Japan, CO₂ emissions fall by over 20%. Stable cities include parts of Western Australia, Russia, and Alaska; while cities like Roebourne in Australia and Baykitskiy in Russia see emissions increase by over 30%, others such as Anchorage in the United States and Gausdal in Norway decrease by over 10%. Among 329 Highly growing cities, 241 are projected to reduce emissions, with Madrid, Rome, Edmonton, Dallas, and Los Angeles all exceeding 20% reductions. Sprawling cities have the highest proportion of emission increases at 31.5%, exemplified by Yenimahalle in Turkey, Yokohama in Japan, and Oulu in Finland, with projected growth of 12.9%, 2.3%, and 11.6%, respectively. Finally, 25.8% of Moderate vertical cities show rising emissions, notably Innsbruck in Austria and Christchurch in New Zealand, which are both expected to grow by over 10%.

We project that under the BAU scenario, although the United States remains the largest emitter among Annex I countries, its total emissions and relative share will both decline. By 2050, its total emissions are projected to reach 4535.3 Mt. This accounts for 37.0% of the total emissions from Annex I countries, down from 39.8% in 2021 (Supplementary Data Fig. 4). Between 2021 and 2050, the United States is projected to reduce its emissions by 1133.4 Mt., contributing to more than half (57.4%) of the total emission reductions. Russia follows closely, with its share of emissions rising from 14.7% in 2021 to 18.2% in 2050. Japan's share in 2050 is 7.6%, followed by Türkiye (4.5%) and Canada (4.4%).

Under the BAU scenario, the average emissions for 46,833 Annex I cities in 2050 are 262,210.6 t, with 2791 cities (6% of the total) dropping below zero. These negative values may indicate cities reaching or moving beyond net zero through future technological advances and energy transitions (DeAngelo et al., 2021). Total emissions for SSP1, BAU, and SSP5 are 11.3, 12.3, and 13.2 Gt, respectively, with Moderate vertical cities accounting for the highest share of emissions across all three scenarios, followed by Intense vertical cities (Fig. 4c–d). Comparing historical and future emission reductions, the historical reduction rate from 2001 to 2020 was 16.9% (Fig. 4e). Among the three future scenarios (2021–2050), only the projected reduction rate for SSP1 (20.5%) exceeds the historical level, while BAU (13.8%) and SSP5 (7.7%) are lower. Moderate vertical cities are the largest contributors to emission reductions. Under the SSP1 scenario, this type has the highest projected reduction rate (22.7%) and contributes 34.5% of the total global emission reductions.

3.4. Per-capita emissions

As shown in the Fig. 5a–c, our projections of per capita CO₂ emissions from 2021 to 2050 reveal significant disparities driven by socioeconomic pathways and urban typologies. As expected, the SSP1 scenario generally promotes the most aggressive decarbonization, achieving the highest reduction percentages across most city types. Conversely, SSP5 consistently shows the lowest mitigation potential, with per capita reductions ranging from 13.6% to 24%.

Among the five urban types, Highly growing and Sprawling cities exhibit substantial decreases in per capita emissions. Highly growing cities achieve the largest overall reduction percentages, falling by 36.3% under the BAU scenario and 34.5% under SSP1. Similarly, Sprawling cities see significant declines, peaking at 31.4% under the SSP1 pathway. On the other hand, Stable cities show minimal progress under the BAU scenario, achieving only a 3.2% reduction by 2050. However, this reduction increases to 25% under the sustainable SSP1 pathway.

In terms of weighted average per capita emissions by 2050, Moderate vertical cities maintain the lowest levels across all pathways, with values of 7, 6.4, and 7.4 t under the BAU, SSP1, and SSP5 scenarios, respectively. In contrast, Stable cities exhibit the highest carbon footprint across all scenarios, peaking at 12.3 t under the BAU scenario. Intense vertical cities maintain moderate and stable levels, with 2050 per capita emissions ranging from 7.4 t (SSP1) to 8.6 t (SSP5).

Among the 42 Annex I countries, the population-weighted mean per capita emission is 8 t, whereas the median is 6 t, suggesting that the overall average is significantly driven by a few high-emitting countries (Fig. 5d). The top 10 countries in terms of per capita emissions account for approximately half of the total population but contribute nearly 70% of the emissions; their per capita emissions are 4–5 times higher than those of the bottom 10 countries.

4. Discussion

Although the spatiotemporal distribution and future projection of urban CO₂ emissions have emerged as critical research topics (Ahn et al., 2023; Albarus et al., 2025; Sun et al., 2023), few studies have leveraged large-scale datasets to explicitly link urban typologies with emission trajectories. This study explores the potential of integrating deep learning with SSPs for city-level CO₂ emission projection, while simultaneously incorporating urban typological classifications. From 2021 to 2050, Annex I cities are projected to reduce total CO₂ emissions by 7.7%–20.5% across the various scenarios. Our results support the hypothesis that emission trajectories vary systematically across and within countries: Moderate vertical cities contribute most to future reductions, whereas Sprawling and Highly growing cities are more associated with continued growth. This heterogeneity indicates that uniform national targets may misrepresent city-level mitigation burdens and supports typology-informed local strategies. Our framework enhances interpretability by providing comparable city-level scenarios and employing typology and feature ablation analyses. Reliability is improved by replacing point estimates with model uncertainty intervals quantified using the Monte Carlo Dropout method.

The IPCC AR6 proposed a classification of urban types based on upward versus outward expansion (Lwasa et al., 2022). Our research extends this framework by incorporating nighttime light intensity, built-up area expansion, and height deviation, employing clustering algorithms for unsupervised classification. The results demonstrate substantial divergence in emission trends across different city types; notably, Intense vertical cities show the most significant trend of rising emissions between 2036 and 2050 (Fig. 3), explaining the observed resurgence in total Annex I emissions during the latter half of the BAU scenario, which likely reflects continued growth eventually outweighing earlier efficiency gains under BAU (Fragkias et al., 2013). Analysis of national emission changes (Supplementary Data Fig. 2) reveals that Russia accounts for 85.5% of the emission growth between 2036 and 2050, whereas nations with continuous reductions, such as the United States and Japan, experience a deceleration in their reduction rates during this period. Russia's contribution may reflect its reliance on fossil fuels and heavy industry. This result is consistent with studies using IAMs (Berrill et al., 2022; Oshiro et al., 2017). We conducted five masking experiments on each projection feature for the country-specific models (Supplementary Data Fig. 3). The results highlight significant disparities in emission projection features across nations: among the five projection features, forest area emerged as the most critical feature for projecting urban emissions in Russia, with its inclusion reducing the projection error by 4.3%. In contrast, population served as the most important feature for emission projections in the United States and Japan. Forest area may indicate land use activity in Russia, while population is more informative in the United States and Japan (Kii, 2019; Uvsh et al., 2020).

We conducted a multi-scale analysis of emission trajectories under the BAU scenario (2021–2050). While total emissions are projected to decrease by 13.8%, the relative rankings of major emitting nations shift slightly (Supplementary Data Fig. 4). We selected the largest emitter and countries with the most significant increases or decreases in emission rankings as representative cases, specifically the United States, Türkiye, and Germany. Notably, although Türkiye's absolute emission growth is modest, its ranking rises from fifth to third due to the decline in emissions across most other nations, while Germany falls from fourth to sixth. At the urban level, in countries with divergent emission trends such as the United States, Türkiye, and Germany, urban emissions exhibit an inverse relationship with the logarithm of their rank, a phenomenon consistent with Zipf's Law (Bettencourt and Zünd, 2020). The partial overlap of the 2021 and 2050 curves indicates a relatively fixed structural hierarchy of urban emissions, where high-emission cities remain the primary contributors relative to lower-emission cities (Cai et al., 2022; Jin and Sharifi, 2025). Beneath this structural stability, however, lies significant heterogeneity in emission distributions. The density curves for the United States and Germany show a distinct leftward shift, with Germany's curve also narrowing, indicating a systematic reduction in urban emissions. Conversely, Türkiye's distribution exhibits the opposite trend; its density curve shifts rightward and broadens by 2050, reflecting an overall increase in emissions.

This research has limitations. The uncertainties mentioned in the introduction, including emissions downscaling, SSP scenario assumptions, and data availability, are still present and will be elaborated below. It focuses on projecting urban emission trajectories based on SSPs and deep learning projection models; however, this approach inherently overlooks technological disruptions and local policy interventions (Riahi et al., 2017). We did not attempt to explicitly model granular adoption of specific low-carbon technologies or the implementation of localized municipal policies, except where such factors are implicitly embedded within historical trends and the narratives of the standardized SSPs (O'Neill et al., 2017). Consequently, should the pace of future technological breakthroughs exceed the baseline assumptions of the SSPs, our projections may overestimate future emissions (Hausfather and Peters, 2020). On the other hand, due to the global lack of consistent city-level or grid-level energy consumption data, our model does not directly incorporate energy mix as a projection feature (Seto et al., 2014). This implies that the model may struggle to fully capture grid decarbonization achieved without concurrent changes in urban typology or economic output, such as a rapid transition from coal to renewables (Creutzig et al., 2015). If a specific city achieves rapid energy transition that is not reflected in socioeconomic indicators, our model may similarly overestimate its emission burden. This limitation is particularly relevant to cities undergoing rapid energy transitions. Future incorporation of measured or bottom-up city-level emission inventories could help improve this framework. Downscaling national emissions with nighttime lights may also underrepresent weakly illuminated sources such as power plants, heavy industry, airports, and transport infrastructure. Consequently, this downscaling approach tends to underestimate industrial- or transport-heavy peripheral cities while overestimating brightly-lit commercial or entertainment centers. Furthermore, our spatial analysis units are based on fixed administrative boundaries rather than dynamically evolving physical built-up areas (Seto et al., 2012). As urban expansion is often fluid, particularly for the Sprawling cities we identified, actual urban activities may spill over administrative borders into surrounding areas (Zhou et al., 2014). By restricting the analysis to static boundaries, we may fail to capture these edge effects, potentially leading to an underestimation of actual urban emissions in these rapidly expanding regions.

While our probabilistic framework effectively captures the epistemic uncertainty within model parameters, it cannot fully account

for the structural uncertainty inherent in the SSP scenarios themselves. Moreover, significant challenges remain regarding global data availability and the interpretability of long-term scenarios (O'Neill et al., 2017). The observed disparities in emission projection features across nations highlight the complexity of localizing global scenarios (Creutzig et al., 2015) (Supplementary Data Fig. 3). Geographic scope represents both a key limitation and a future priority. Due to data reliability constraints, this research focused on Annex I nations; however, the majority of future urbanization and potential emission growth is expected to occur in the Global South (non-Annex I countries). Extending our projection framework to these rapidly developing regions is essential for completing the global picture (Nagendra et al., 2018). Additionally, future research should aim to integrate data-driven approaches with process-based energy models to better constrain the physical limits of decarbonization (Gurney et al., 2015).

5. Conclusion

This study developed a city-level framework combining urban typology classification and deep learning to project CO₂ emissions for 46,833 Annex I cities under BAU, SSP1, and SSP5 through 2050. By integrating socioeconomic projections with Monte Carlo Dropout, the framework captures differences among cities and quantifies uncertainty in their future emission trajectories.

Total emissions are projected to decline by 20.5% under SSP1, 13.8% under BAU, and 7.7% under SSP5 between 2021 and 2050, although reductions slow or reverse after 2036 in some scenarios and urban types. Moderate vertical cities make the largest contribution to emission reductions and, under SSP1, achieve a 22.7% reduction while accounting for 34.5% of the total reduction. Per capita results also show marked differences, with Moderate vertical cities maintaining the lowest levels and Stable cities the highest across the three scenarios.

These results show that national trends conceal substantial variation among cities. Linking urban development patterns with scenario-based projections provides a more detailed basis for understanding the distribution of future urban emissions across countries and city types.

CRedit authorship contribution statement

Yukai Jin: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ayyoob Sharifi:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Code availability

The code of the proposed model is publicly available on GitHub: <https://github.com/UrbanModeling/FiLM-Seq2Seq>.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: (Given his role as the associate editor of the journal, Ayyoob Sharifi had no involvement in the peer-review of this article and has no access to information regarding its peer-review. Full responsibility for the editorial process for this article was delegated to another journal editor. Yukai Jin declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.)

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Claude and Grammarly to check the grammar, punctuation, and clarity. After using these tools, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.uclim.2026.103160>.

Data availability

All data used in this study are publicly accessible. CO₂ emission time-series data for Annex I countries were collected from the UNFCCC Greenhouse Gas Inventory Data Interface (Kuyper et al., 2018). Population data were obtained from the Gridded Population of the World dataset (Doxsey-Whitfield et al., 2015), and GDP data were sourced from the global gridded gross domestic product (GDP) dataset by Kummu et al. (2025). Nighttime light data were retrieved from the Annual Simulated VIIRS Nighttime Light Dataset by Chen et al. (2024), and SSP scenario data were collected from the IIASA SSP Database (Riahi et al., 2017) (Version 2.0). Boundary data were obtained from Global Administrative Areas (GADM, 2018) (GADM, version 4.1) (<https://gadm.org>). Land cover data (MCD12Q1 V6.1) are available through the NASA LP DAAC (Friedl & Sulla-Menashe, 2022), and global urban building height data are available from the repository associated with Y. Y. Zhou et al. (2022).

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